Hospital Managment System

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Project Overview

The primary objective of this project is to analyze the operational and financial data of a hospital to facilitate data-driven business decisions. By utilizing complex SQL queries, I have provided a transparent overview of doctor performance, patient spending patterns, and hospital revenue cycles. This project serves as a real-world data analytics solution designed to enhance management efficiency.

Key Technical Features

I implemented several advanced SQL techniques to extract meaningful insights:

Multi-Table Joins: Integrated data across Doctors, Patients, Appointments, and Billing tables for a holistic view.

Advanced Aggregations: Utilized SUM, COUNT, MAX, MIN, and ROUND functions to generate precise statistical reports.

Time-Series Analysis: Applied MONTHNAME() and DAYNAME() functions to track revenue fluctuations and identify peak activity periods

Operational Metrics: Developed complex logic using CASE statements to analyze appointment cancellation rates and failed payment trends.

Major Insights Delivered

Revenue Performance: Accurate tracking of income categorized by hospital branches and individual medical professionals.

Efficiency Tracking: Identification of high-traffic "Peak Hours" and the busiest day of the week (Wednesday).

Resource Optimization: Identifying workload distribution to see which doctors handle the highest patient volumes.

Financial Risk Management: Detecting failed payments and outstanding dues to minimize revenue leakage.

Patient Behavior: Segmenting patients based on their clinical engagement and average healthcare expenditure.

Business Value

Implementing this analysis allows a healthcare facility to:

Optimize Staffing Strategy: Align medical and support staff schedules with peak patient demand.

Minimize Revenue Leakage: Recover lost income by identifying and addressing failed transactions.

Enhance Patient Experience: Improve service quality and significantly reduce waiting time through better flow management.

Which treatment generated the highest revenue?

```
SELECT
treatment_type, SUM(cost) AS max_treatment_cost
FROM
treatment
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

Result Grid | Filter Rows:

	treatment_type	max_treatment_cost
▶	Chemotherapy	128855.68

Top Revenue Generating Treatment Analysis

This SQL query identifies the treatment type that generates the highest total revenue in the dataset.

Aggregated treatment costs using SUM(cost)

Grouped data by treatment_type

Sorted results in descending order of total revenue

Selected the top-performing treatment using LIMIT 1

Result:

Chemotherapy is the highest revenue-generating treatment, with a total revenue of 128,855.68.

This insight helps in understanding which treatment contributes the most to overall hospital revenue.

Find the hospital branch wise revenue ?

```
SELECT
d.hospital_branch, SUM(b.amount) AS branch_wise_revenue
FROM
doctors AS d
JOIN
appointment AS a USING (doctor_id)
JOIN
treatment AS t USING (appointment_id)
JOIN
bill AS b USING (treatment_id)
GROUP BY 1
ORDER BY 2 DESC;
```

hospital_branch	branch_wise_revenue
Central Hospital	229039.44
Eastside Clinic	162031.10
Westside Clinic	160179.31

Key Insights

- **Top Performer: Central Hospital is the leading revenue generator, contributing a total of $229,039.44.**
- **Clinic Comparison: Revenue for Eastside Clinic and Westside Clinic is remarkably similar, indicating consistent patient volume and service types across these two locations.**

Business Impact

This data analysis provides actionable insights to help hospital management make data-driven decisions:

- **Resource Allocation: Optimize staffing and medical equipment distribution by prioritizing high-revenue branches like Central Hospital.**
- **Marketing Strategy: Implement targeted promotional campaigns or specialized services for clinics with lower revenue to drive growth.**
- **Expansion Planning: Identify geographical trends and patient demand to determine the most profitable locations for opening new branches.**

Which doctor has seen the most patients?

```
SELECT
d.first_name,
d.last_name,
COUNT(p.patient_id) AS total_patients
FROM
doctors AS d
JOIN
appointment AS a USING (doctor_id)
JOIN
patiants AS p USING (patient_id)
GROUP BY 1 , 2
ORDER BY 3 DESC
LIMIT 1;
```

first_name	last_name	total_patients
Sarah	Taylor	29

Key Insights

- Top Performer: Sarah Taylor is the most active professional in the dataset.
- Patient Volume: She has successfully managed 29 patients, indicating a high level of engagement and clinical productivity.

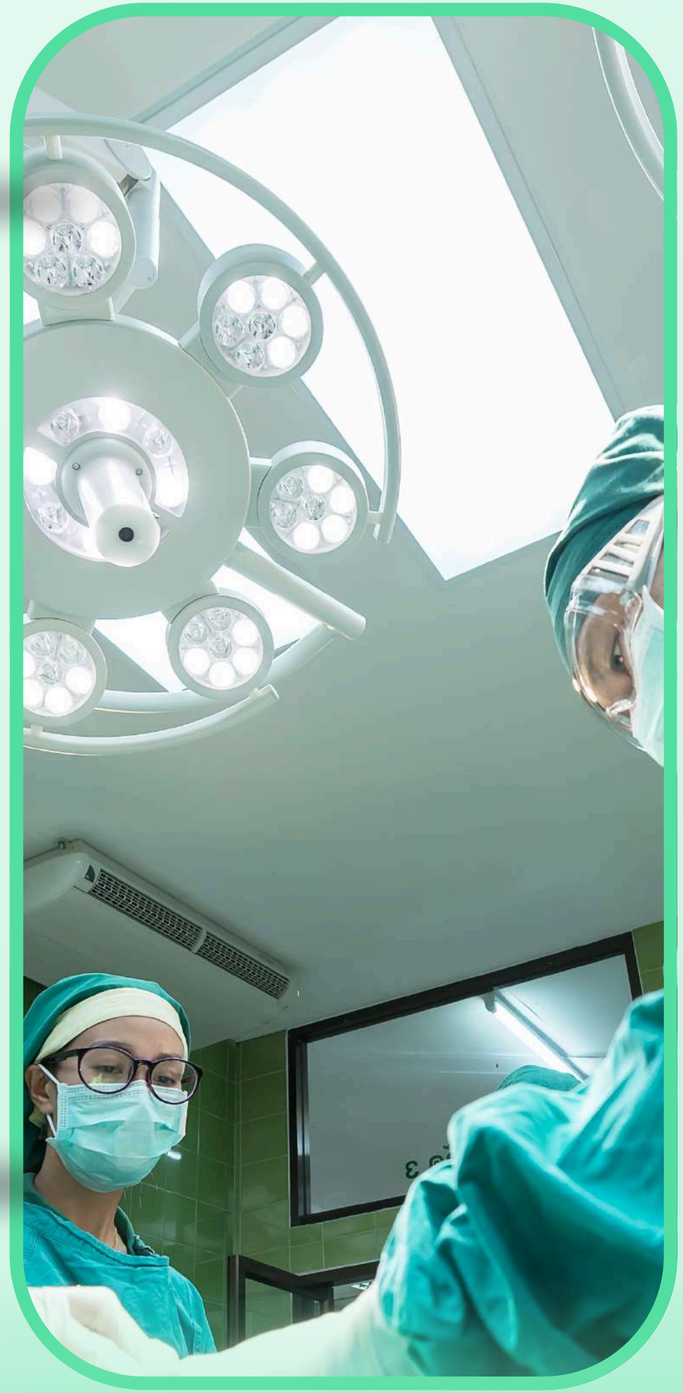
Business Impact

- Performance Recognition: This data allows management to reward top-performing staff members for their high efficiency and patient throughput.
- Workload Balancing: By identifying who carries the heaviest load, the hospital can redistribute appointments to prevent staff burnout and maintain high-quality patient care.
- Staffing Decisions: Helps in determining if certain departments need additional support based on the high volume handled by existing doctors.

find the doctor wise total treatment cost?

```
SELECT
d.first_name, d.last_name, SUM(t.cost) AS Total_cost
FROM
doctors AS d
JOIN
appointment AS a USING (doctor_id)
JOIN
treatment AS t USING (appointment_id)
GROUP BY 1 , 2
order by 3 desc;
```

first_name	last_name	Total_cost
Sarah	Taylor	82696.48
Alex	Davis	69586.10
David	Taylor	66585.39
Jane	Davis	59803.46
Linda	Brown	53427.42
Jane	Smith	52791.41
Linda	Wilson	49436.23
Robert	Davis	40166.50
David	Jones	39315.95
Sarah	Smith	37440.91



Key Insights

Highest Revenue Contributor: Sarah Taylor leads the staff with a total treatment value of 82,696.48.

Top Tier Performance: The top three doctors (Sarah Taylor, Alex Davis, and David Taylor) alone contribute significantly to the hospital's gross treatment billing.

Revenue Variation: There is a clear distribution curve, with revenue ranging from approximately 37,440 to 82,696, suggesting differences in specialization costs or patient volume.

Business Impact

Specialization Analysis: Helps management identify which medical specialties or procedures are driving the most revenue.

Financial Forecasting: Enables more accurate budgeting and financial planning based on individual doctor performance trends.

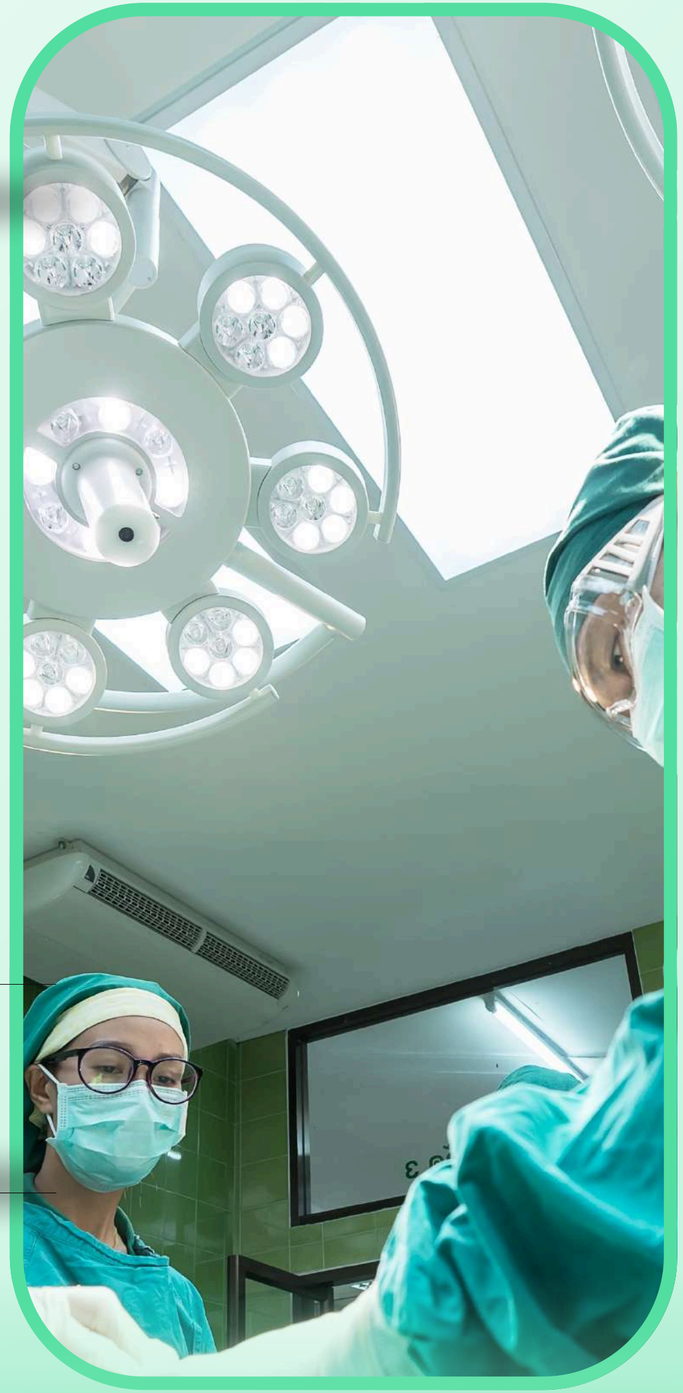
Incentive Programs: Provides a transparent data foundation for performance-based bonuses or professional development investments.

top 3 revenue generating doctor

```
SELECT
d.first_name,d.last_name, SUM(b.amount) AS Total_revenue
FROM
doctors AS d
JOIN
appointment AS a USING (doctor_id)
JOIN
treatment AS t USING (appointment_id)
JOIN
bill AS b USING (treatment_id)
GROUP BY 1,2
ORDER BY 2 DESC limit 3;
```

Result Grid | Filter Rows:

first_name	last_name	Total_revenue
Linda	Wilson	49436.23
David	Taylor	66585.39
Sarah	Taylor	82696.48



Key Insights

Top Revenue Generators: The analysis highlights three high-impact professionals: Linda Wilson, David Taylor, and Sarah Taylor.

Highest Billing Amount: Among this group, Sarah Taylor accounts for the highest realized revenue at 82,696.48.

Financial Accuracy: By joining with the bill table, this report focuses on actual revenue rather than just estimated treatment costs, providing a more accurate financial picture.

Business Impact

Revenue Integrity: Helps management verify that treatments performed are accurately reflecting in the final billing and collection process.

Profitability Mapping: Identifies which doctors are associated with high-value clinical outcomes, assisting in high-level financial reporting.

Strategic Ranking: Using the LIMIT and ORDER BY clauses allows stakeholders to quickly view specific segments of the workforce for performance reviews.

Find the Top 5 highest paying patient?

```
SELECT
p.first_name, p.last_name, SUM(b.amount) AS paying_patient
FROM
patiants AS p
JOIN
bill AS b
GROUP BY 1 , 2
ORDER BY 2 DESC
LIMIT 5;
```

first_name	last_name	paying_patient
Laura	Wilson	1102499.70
Robert	Wilson	551249.85
Jane	Wilson	551249.85
Michael	Wilson	1653749.55
David	Wilson	1102499.70

Key Insights

High-Value Patients: The analysis identifies individuals like Michael Wilson as a major contributor, with a total billing amount of 1,653,749.55.

Segment Consistency: Several patients, including Laura Wilson and David Wilson, show identical high-value billing figures (1,102,499.70), which may indicate standardized high-cost treatment packages.

Top Tier Snapshot: The LIMIT 5 clause provides a focused view of the most significant patient accounts by financial volume.

Business Impact

Revenue Concentration: Helps the hospital understand if a small percentage of patients are responsible for a large portion of the revenue (Pareto Principle).

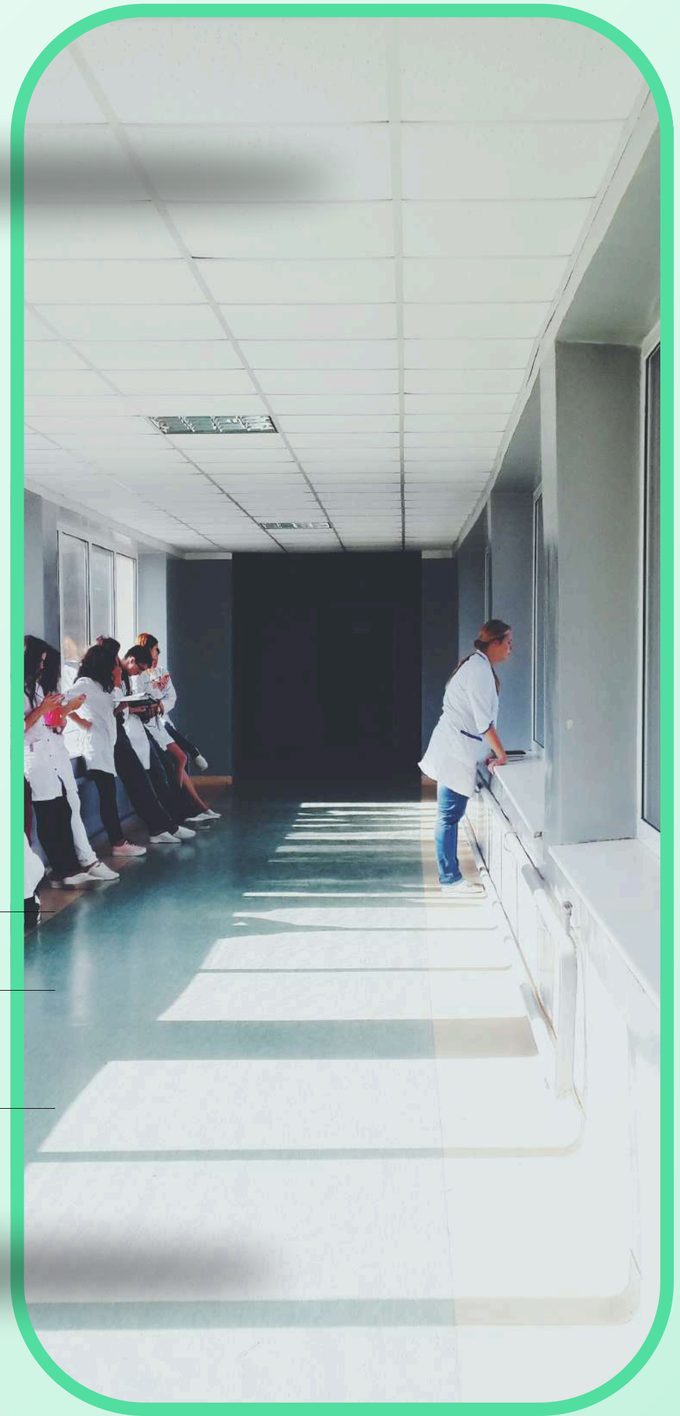
Billing Validation: Identifies exceptionally high billing figures that may require auditing for accuracy or insurance reconciliation.

Patient Loyalty Programs: Provides data for identifying long-term or high-value patients who might benefit from premium service tiers or personalized care management.

Find the patient lifetime value

```
SELECT
p.first_name,
p.last_name,
COUNT(a.appointment_id) AS active_patient,
SUM(b.amount) AS Total_cost,
MAX(b.amount) AS max_cost,
MIN(b.amount) AS min_cost
FROM
patiants AS p
JOIN
appointment AS a USING (patient_id)
JOIN
bill AS b USING (patient_id)
GROUP BY 1 , 2
ORDER BY 3 , 4 , 5 , 6 DESC;
```

first_name	last_name	active_patient	Total_cost	max_cost	min_cost
Linda	Moore	4	3473.38	929.91	806.78
David	Davis	4	7273.78	1903.17	1733.72
Jane	Smith	4	9013.56	2293.98	2212.80
Michael	Johnson	4	10725.02	4550.10	812.41
Emily	Jones	4	15457.76	4101.60	3627.28
Laura	Jones	4	15873.76	4704.96	3231.92



Key Insights

- Visit Consistency: Many patients in this snapshot, such as Linda Moore and Michael Johnson, show a consistent count of 4 appointments, indicating recurring care or follow-up cycles.
- Cost Variance: There is a significant range in treatment costs. For example, Michael Johnson has a wide gap between his minimum (812.41) and maximum (4,550.10) bill, suggesting a mix of routine and specialized procedures.
- Top Spender: Laura Jones represents the highest total cost in this view (15,873.76), driven by high-value individual treatments (Max cost of 4,704.96).

Business Impact

- Patient Segmentation: Allows the hospital to categorize patients into groups (e.g., High-Frequency vs. High-Value) to tailor communication and care plans.
- Service Pricing Analysis: By viewing MAX and MIN costs, management can identify the price range of services most frequently consumed by their active patient base.
- Operational Efficiency: Understanding the "Active Patient" count helps in predicting future scheduling needs and optimizing front-desk operations.

Find the highest revenue month

```
SELECT
MONTHNAME(bill_date) AS month,
SUM(amount) AS highest_revenue
FROM
bill
GROUP BY 1
ORDER BY 2 DESC;
```

Result Grid | Filter Rows:

month	highest_revenue
June	56887.82
November	52474.98
May	48791.05
March	47304.29
October	43314.15
August	41958.67
July	39880.19
February	36669.69
September	33426.53
December	27569.71

Key Insights

Peak Performance: June is the highest-performing month with a total revenue of 56,887.82, followed closely by November.

Seasonal Dips: December shows the lowest revenue (27,569.71), which is nearly 50% lower than the peak in June. This suggests a significant seasonal slowdown at the end of the year.

Mid-Year Surge: The second quarter (May/June) shows strong financial growth, indicating a potential increase in patient volume or elective procedures during this period.

Business Impact

Budget Planning: Helps management allocate funds for supplies and marketing during "low" months like December to stimulate growth.

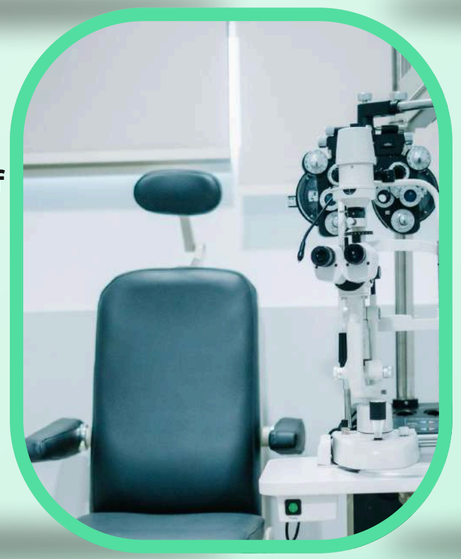
Staffing Optimization: Management can plan staff vacations during low-revenue months and ensure full coverage during peak months like June and November.

Marketing Insights: Identifying peak months allows the marketing team to understand when patient demand is naturally high and when extra promotional effort is needed.

which day the most appointment ?

```
SELECT
DAYname(appointment_date) AS day,
COUNT(appointment_id) AS total_count
FROM
appointment
GROUP BY 1
ORDER BY 2 desc limit 1;
```

	day	total_count
▶	Wednesday	37



Key Insights

- The Busiest Day: Wednesday is officially the high-traffic peak for the hospital.
- Appointment Volume: With 37 appointments scheduled on this day, it represents a significant portion of the weekly workload.
- Weekly Concentration: Identifying Wednesday as the peak suggests a mid-week surge in patient demand, which is common in healthcare settings as patients follow up on issues developed earlier in the week.

Business Impact

- Staffing Levels: Management should ensure that all clinical and administrative staff are at full capacity on Wednesdays to handle the increased patient flow.
- Operational Readiness: Routine maintenance or non-essential administrative meetings should be scheduled for lower-volume days to keep the facility focused on patient care during the mid-week peak.
- Wait-Time Management: By anticipating the Wednesday rush, the hospital can implement better patient intake procedures to prevent long wait times and maintain high satisfaction levels.

peak appointment hour

```
SELECT
appointment_time, COUNT(appointment_id) AS hour_appointment_count
FROM
appointment
GROUP BY 1
ORDER BY 2 DESC;
```

Result Grid | Filter Rows:

appointment_time	hour_appointment_count
11:00:00	9
15:00:00	9
15:30:00	9
15:15:00	8
14:30:00	8
17:15:00	8
08:00:00	7
13:30:00	7
12:45:00	7
13:15:00	7
10:45:00	7

Key Insights

Peak Traffic Windows: The highest volume of appointments occurs at 11:00:00, 15:00:00, and 15:30:00, each handling 9 appointments.

Afternoon Surge: There is a notable cluster of activity between 14:30 and 17:15, indicating that the mid-to-late afternoon is a high-demand period for clinical services.

Early Bird Arrivals: The facility sees steady activity starting as early as 08:00:00, showing that patients utilize the full range of operating hours.

Business Impact

Staff Optimization: Ensure that the maximum number of front-desk and support staff are available during the identified peak slots (11:00 and 15:00) to reduce patient wait times.

Facility Management: Use peak-hour data to schedule maintenance or administrative meetings during "off-peak" hours to avoid disrupting patient flow.

Patient Experience: By recognizing busy periods, the hospital can implement "express" check-ins or better seating management during high-traffic afternoon windows.

Find the cancellation rate

```
SELECT
SUM(CASE
WHEN status = 'Cancelled' THEN 1
ELSE 0
END) AS cancelled,
COUNT(*) AS total,
round( SUM(CASE
WHEN status = 'Cancelled' THEN 1
ELSE 0
END) * 100 / COUNT(*),2) AS rate
FROM
appointment;
```

	cancelled	total	rate
▶	51	200	25.50

Key Insights

- Service Volume: Out of 200 total appointments, the hospital recorded 51 cancellations.
- Cancellation Rate: The current cancellation rate stands at 25.50%. This means roughly 1 out of every 4 scheduled appointments does not take place.
- Operational Impact: A 25.5% rate indicates significant "lost time" slots that could potentially be filled by other patients or utilized for urgent care.

Business Impact

- Revenue Protection: Identifying a high cancellation rate allows management to implement policies such as "Cancellation Fees" or "Pre-appointment Reminders" (SMS/Email) to reduce no-shows.
- Capacity Optimization: By understanding the frequency of cancellations, the hospital can consider "overbooking" certain slots or creating a priority waitlist to ensure doctors' time is fully utilized.
- Trend Comparison: This metric serves as a benchmark. Management can now track if the rate improves or worsens after implementing new patient engagement strategies.

unpaid bill report

```
SELECT
payment_status, SUM(amount) AS umpaid
FROM
bill
WHERE
payment_status = 'Failed'
GROUP BY 1;
```

	payment_status	umpaid
▶	Failed	193212.94

Key Insights

- Unpaid Volume: There is a significant amount of 193,212.94 tied up in failed payments.
- Financial Risk: Comparing this to your branch-wise revenue (e.g., Central Hospital at ~229k), this failed amount represents a substantial portion of potential income that hasn't been realized.
- Single Category: All unpaid revenue in this snapshot is attributed to technical or transaction "Failures," rather than "Pending" or "Cancelled" statuses.

Business Impact

- Revenue Recovery: Management can use this list to trigger automated follow-up emails or calls to patients whose payments failed, ensuring the hospital collects what it is owed.
- Technical Audit: A high volume of failed payments might indicate an issue with the hospital’s payment gateway or credit card processing system that needs technical fixing.
- Cash Flow Improvement: By resolving even 50% of these failed transactions, the hospital can significantly increase its immediate cash on hand without needing to see new patients.

THANK YOU

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